

University of Botswana
Department of Computer Science

CSI247: Data Structures
Course Outline: 2026/27 (Sem I)

Lecturer	Mrs B. Gopolang
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Consultation Hours	M, W & F, 8 – 12pm (Except M, 11-12). Additionally, R 7-1pm. Email or WhatsApp appointments are welcome if you find these times unfavourable.

1. Course Objectives:

In this course, students will:

- Learn about some of the common data structures.
- Learn about common applications for data structures discussed in the course
- Implement these common data structures using a high-level language (Java)
- Write programs using discussed data structure
- Choose appropriate data structures for any given problem
- Understand costs and benefits associated with these data structures.

2. Pre-requisite course

- CS students: CSI132 and CSI142
- IS students: ISS112 ($\geq 50\%$)

3. Timetable

a) Lectures: M, W & F: 7am – 8am, 252/001

b) Labs: R, 7-9, 9-11am or 11-1pm

- Pick one and maintain it throughout the semester
- Lab registration will be done online via CS Moodle

Note: Attendance for lectures, labs and tutorials is **COMPULSORY!**

4. Topics to be covered

Topics will include, but not limited to, the following:

- a) Review of Java programming basic concepts, including User-defined classes and arrays
- b) Recursion
- c) Big-O notation
- d) Searching and sorting algorithms (using arrays)
- e) Packages
- f) Object-oriented programming
- g) Generics
- h) Using Java pre-defined common data structures
- i) Implementation of common data structures
 - Linked lists
 - Stacks
 - Queues
 - Hash tables
 - Trees
 - Graphs

5. Course Material

Lecture notes will be posted on CS Moodle (<http://www.cs.ub.bw/moodle>) before each lesson. You MUST always go through them before the lesson.

6. Recommended Readings

Data Structures & Algorithms in Java by Goodrich, Tamassia & Goldwasser, 6th Ed, 2014

7. Assessment Criteria

- 2 tests (30%)
- 4 quizzes (20%)
 - Refer to Moodle for dates and times for tests and quizzes
- Final exam (50%)

Academic Dishonesty

- Refer to UB's academic calendar to familiarize yourselves with the relevant academic rules and regulations.
- Take note that the university has an Academic Dishonesty Policy, and appropriate action will be taken where necessary